UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

ILLINOIS CyberGIS Center FOR ADVANCED DIGITAL AND SPATIAL STUDIES

Floability

Interactive, Reproducible Notebook Workflows at Scale

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CyberGIS Center for Advanced Digital and Spatial Studies

Research & Innovation Forum (RIF)

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Notebook Workflows

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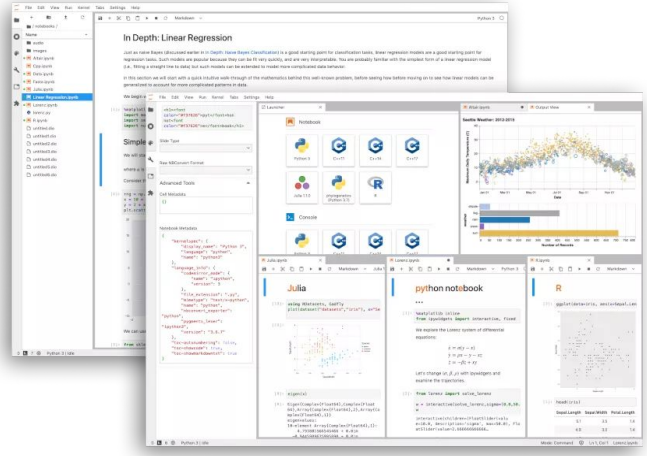
Interactive, Graphical, Unified

- Increasingly popular among domain experts
- Good for initial exploration
- Interactive
- Document with code
- Reproducible & Shareable



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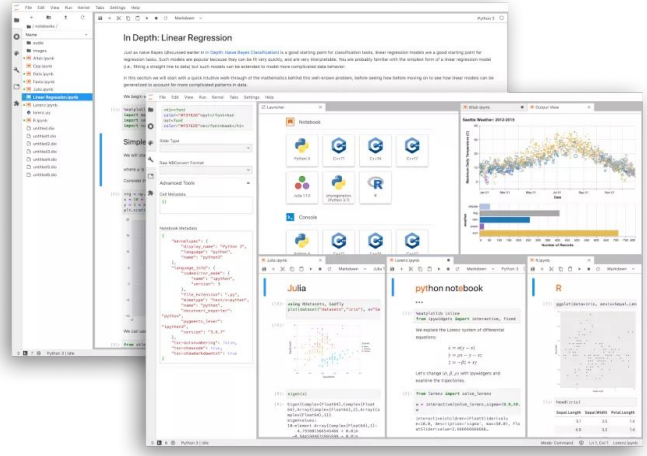
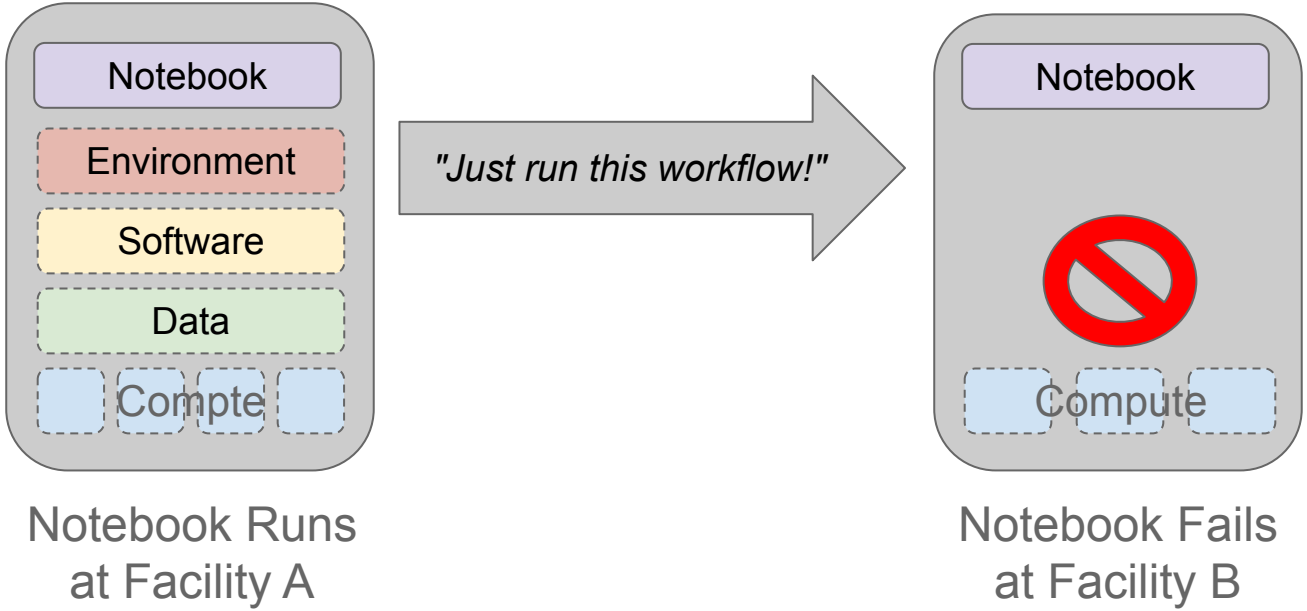
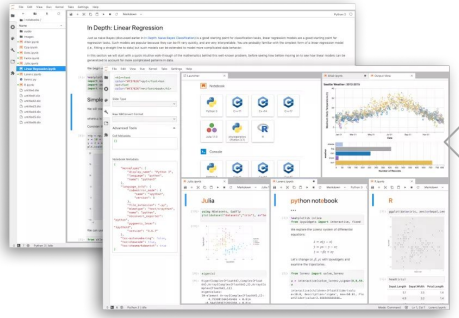
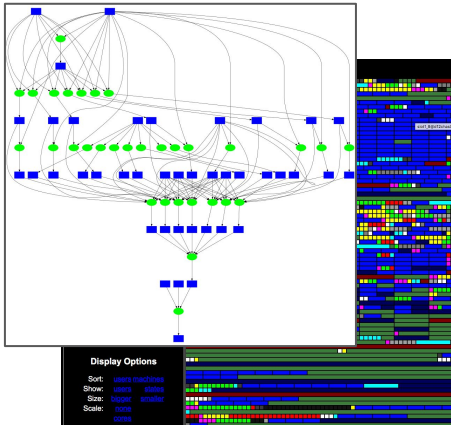


Challenge


Notebook
Environment
Software
Data
Compte
"Just run this workflow!"
Notebook
Compute
Notebook Runs
at Facility A
Notebook Fails
at Facility B

The **Floability Project** aims to enable the rapid and portable deployment of notebooks expressing complex scientific workflows across a wide range of cyberinfrastructure.





Floability Project Scope

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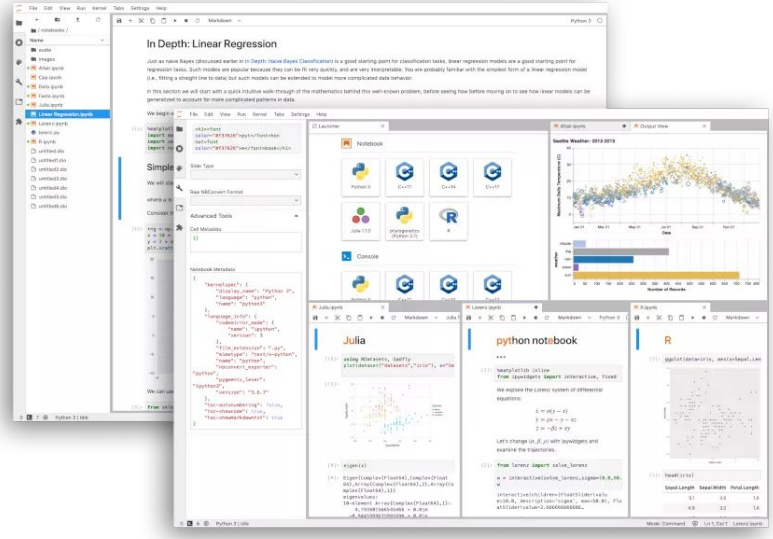
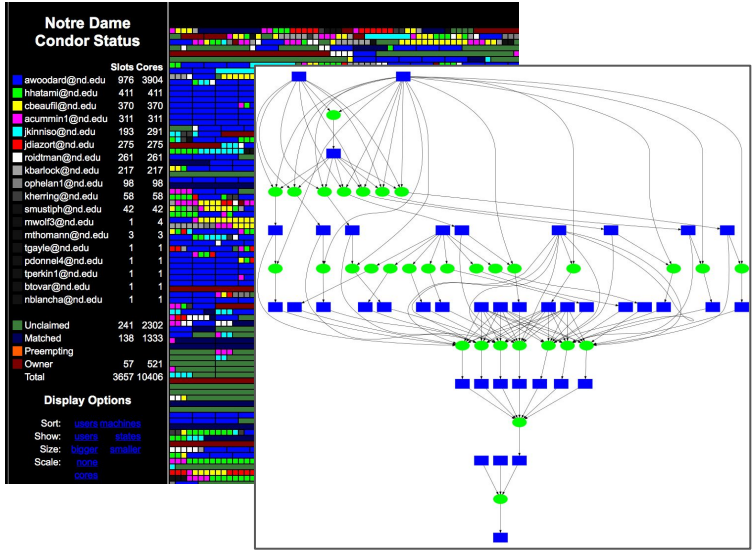
Interactive Notebooks

Graphical, Interactive, Personal, **Limited**



High Throughput Workflows

Scripting, Batch, Shared, **Scalable**



Software Hacking Scope

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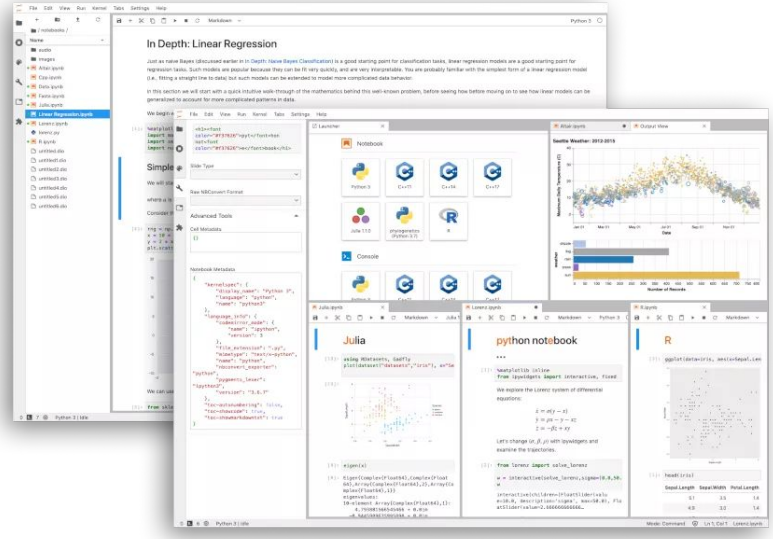
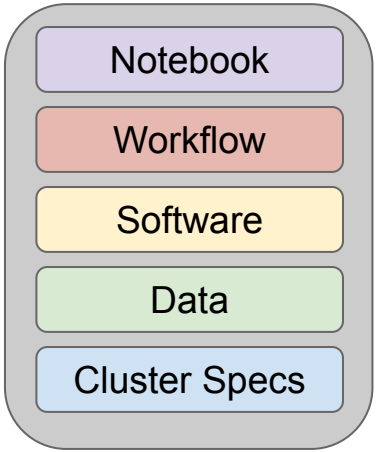


High Throughput Workflows

Scripting, Batch, Shared, **Scalable**

Backpack

A **Backpack** contains everything required to deploy a notebook workflow



A backpack encapsulates a notebook workflow with structured specifications of all its dependencies—data, software,and computational resources—needed for seamless execution across diverse environments.

Demo - Setup!

- Prerequisites
  - Python
  - Conda/mamba

- Install floability

```
conda install -c conda-forge floability
```

- Download floability example backpacks

```
git clone https://github.com/floability/floability-examples.git
```

Demo - Run!

- Run vulnerability analysis for aging dams workflow

```
floability run --backpack
floability-examples/iguide-gis-aging-dams --no-worker
```

- Access Jupyter interface on provided link

```
http://localhost:8888/lab/?token=5560dfc97171845ad1dada4cb7d2…
```

Demo - Run!

- Run vulnerability analysis for aging dams workflow

```
floability run --backpack
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```

- Access Jupyter interface on provided link

```
http://localhost:8888/lab/?token=5560dfc97171845ad1dada4cb7d2…
```

- A few notes
  - Fix data path in Cell# 2
  - Ignore errors (Spark internal errors + CSRF etc.)
  - You're good as long as you see output graphs and tables

What Exactly is a Backpack?

- ➔ workflow
  - ◆ Local_Analysis_Multi_Scenario.ipynb
- ➔ software
  - ◆ environment.yml
- ➔ data
  - ◆ data.yml
  - ◆ dam_list.csv
  - ◆ census_geometry
  - ◆ NID_FIM_TAS_F

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```
ef analysis(events):
    warnings.filterwarnings("ignore", module="coffea.*"

    dataset = events.metadata["dataset"]
    events["PFCands", "pt"] = events.PFCands.pt * event

    cut_to_fix_softdrop = ak.num(events.FatJet.constitu
    events = events[ak.all(cut_to_fix_softdrop, axis=1)

    trigger = ak.zeros_like(ak.firsts(events.FatJet.pt)
    for t in triggers["2017"]:
        if t in events.HLT.fields:
            trigger = trigger | events.HLT[t]
    trigger = ak.fill_none(trigger, False)

    events["FatJet", "num_fatjets"] = ak.num(events.Fat

    goodmuon = (
        (events.Muon.pt > 10)
        & (abs(events.Muon.eta) < 2.4)
        & (events.Muon.pfRelIso04_all < 0.25)
        & events.Muon.looseId
    )

    nmuons = ak.sum(goodmuon, axis=1)
```

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```yaml
name: my_mdv5_env

channels:
  - conda-forge

dependencies:
  - python=3.12
  - cloudpickle
  - ndcctools
  - coffea<2024.4.2
  - dask<2024.5.2
  - dask-awkward<2024.5.0
  - dask-core<2024.5.0
  - dask-histogram<2024.5.0
  - fsspec
  - pip:
    - fastjet<3.4.2.2
```

What Exactly is a Backpack?

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- workflow
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  - NID_FIM_TAS_F

```yaml
data:
  - name: "census_data"
    source_type: "backpack"
    source: "data/census_geometry/census_data.csv"
    target_location: "data/census_geometry/census_data.csv"

  - name: "census_tract_from_api"
    source_type: "backpack"
    source: "data/census_geometry/census_tract_from_api.geojson"
    target_location: "data/census_geometry/census_tract_from_api.geojson"

  - name: "state_lookup"
    source_type: "backpack"
    source: "data/census_geometry/state_lookup.csv"
    target_location: "data/census_geometry/state_lookup.csv"
```

Thank You!

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Comments / Questions?

https://floability.github.io/